Zepto Quick Commerce Data Analysis

Delivery Time, Order Patterns, and Product Demand Trends

Tracking Best-Sellers, SLA Risk, Peak Ordering Windows, Zone Demand & Inventory Risk

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Report type	Descriptive analytics (What happened?)

Note on Data

Live Zepto order data is not publicly accessible, so this report runs on a synthetic dataset built to mirror real quick-commerce order patterns:

Zepto Quick Commerce Data Analysis

10 products across 6 categories, 5 dark-store zones, over a 15-day window (Jul 01, 2026 – Jul 15, 2026), for 150 product-day records in total.

Each row represents one product's aggregated activity for one day at its dominant dark-store zone — orders, average delivery time, revenue, customer rating, the time-of-day band with the heaviest ordering, and remaining stock level. This is the same granularity a real quick-commerce analytics pipeline would expose at the reporting layer: raw per-order events are rolled up before they ever reach a dashboard (see Section 3).

- one product (Amul Toned Milk) generates disproportionate order volume, mirroring the Pareto pattern common on quickcommerce apps' daily-essentials lines.
- Weekend and evening demand surges layered on top of small categorylevel growth trends.
- A slow-building SLA risk signal at one dark-store zone, where average delivery time climbs past the 10-minute target over the window — the kind of trend an operations dashboard should surface early.
- Uneven zone-level demand concentration, and two products deliberately depleting into Low Stock / Out of Stock states.

Objective

Build a descriptive analytics layer over daily Zepto order data that answers five recurring, operational business questions — the kind a category manager, dark-store operations lead, or delivery-ops team would otherwise have to ask an analyst for every morning:

#	
Q1	What are the best-selling products, and by how much do they lead the pack?
Q2	How are delivery times trending, and which dark-store zone is at risk of breaching the 10-minute SLA?
Q3	When do customers order most — by time of day and day of week?
Q4	Where is order demand concentrated across dark-store zones, and what does that imply for fulfillment?
Q5	Which products are at risk of stocking out at the dark stores, given current order velocity?

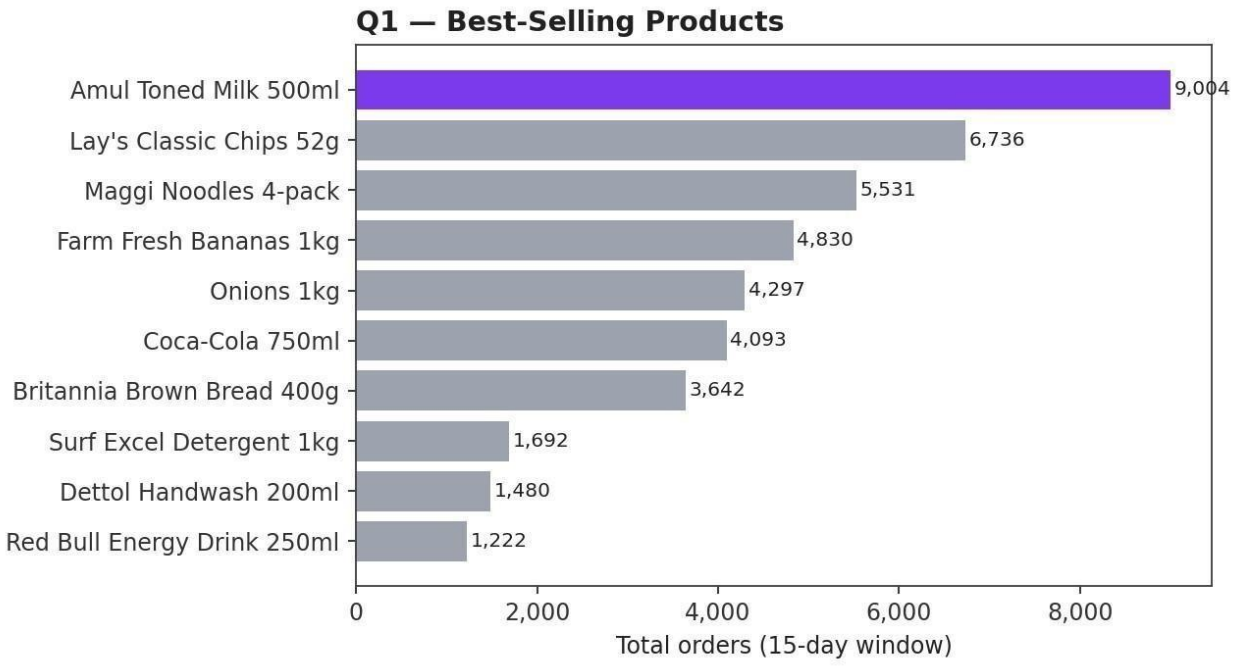
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How This Would Work at Real Scale

Processing millions of Zepto orders per day across hundreds of dark stores is an engineering problem before it is an analytics one. A realistic architecture behind a report like this one:

Layer	Purpose	Typical Tools
Investigation	Capture each order event (placed, packed, dispatched, delivered) in real-time	CSV dataset / Order log
Stream processing	Roll raw events into per-product, per-zone, per-minute aggregates; flag late deliveries	Pandas
Storage	Hold raw events + pre-aggregated tables	CSV / Excel files
Batch aggregation	Nightly rollups — daily orders, delivery-time averages, inventory reconciliation	Pandas, NumPy
Serving / BI layer	Dashboards for dark-store ops and category teams	Matplotlib / Seaborn

I. Best-Selling Products



- **Amul Toned Milk 500ml** led with **9,004 orders**, 33.7% ahead of the second-place product (**Lay's Classic Chips 52g**, 6,736 orders).

- The top SKU alone accounts for **21.2%** of all orders across 10 products — a classic *hero-SKU* pattern where a small number of daily-essentials listings drive a disproportionate share of order volume.

- Ranking logic here is orders, not revenue: a low-price staple like milk outselling a premium item in order count matters differently to dark-store slotting (pick/pack load, shelf placement near the entrance) than it does to finance (margin contribution) — a real dashboard should expose both cuts as separate views.

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The analysis ranks the products based on their total number of orders over the 15-day period. Amul Toned Milk 500ml was the best-selling product with 9,004 orders, followed by Lay's Classic Chips 52g with 6,736 orders. Amul Toned Milk recorded 33.7% more orders than the secondbest-selling product.

The top-selling product accounted for 21.2% of the total orders across all
10 products, showing a Hero-SKU effect, where a single product contributes a disproportionately large share of the total order volume.

This pattern is related to the Pareto Principle (80/20 Rule), where a small number of products can generate a large portion of overall demand. Therefore, high-demand products should receive greater attention in stock management, shelf placement, replenishment, and fulfillment operations.

With Amul Toned Milk 500ml carrying 21.2% of total orders, it should get priority treatment on three fronts: it should be the first SKU checked in every dark store's stock audit (a stockout here costs more than on any other SKU), it should sit in the fastest-pick shelf position, and its delivery-time performance (Section 5) should be tracked separately from the zone average — a slow hero SKU delivery disproportionately damages the habitual 10-minute-delivery expectation that keeps customers coming back.

KPI definitions used:

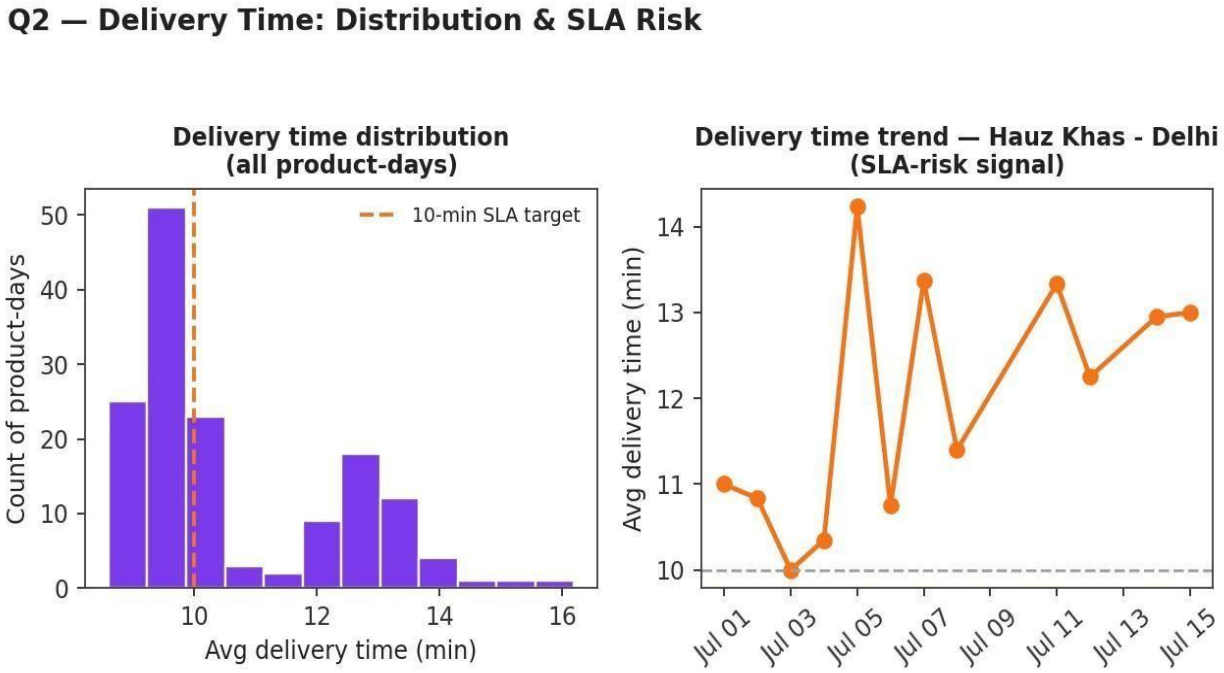
Total Orders = Sum of orders across all products during the analysis period.

Top Product Share = Top-selling product orders ÷ total orders × 100.

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II. Delivery Time: Distribution & SLA Risk

Left: distribution of average delivery times across all 150 product-day records against the 10-minute SLA target.
Right: delivery-time trend at the zone showing the clearest upward drift.



Generated with pandas (aggregation) + matplotlib (visualization) directly from zepto_order_data.csv.

- Average delivery time across the window is 10.7 minutes, but 40.0% of product-days already exceed the 10-minute target — the SLA is being missed often enough to matter, not just on rare outlier days.

- Hauz Khas - Delhi is the clearest risk: average delivery time there rose from 11.0 min to 13.0 min over the window — a steady climb rather

  than a single bad day, which is the earliest signal of a darkstore capacity or rider-availability issue worth escalating before it shows up in churn.

- In production this view is most useful segmented, not aggregated: zones trending upward in rolling 3-day delivery time, and zones combining high order volume with SLA breaches, are the two highestpriority slices — volume amplifies the customer-facing damage of a slow zone.

The overall average delivery time was 10.7 minutes, compared with the 10-minute SLA target. 40% of product-days exceeded the 10-minute SLA, showing that delivery-time breaches were occurring frequently.

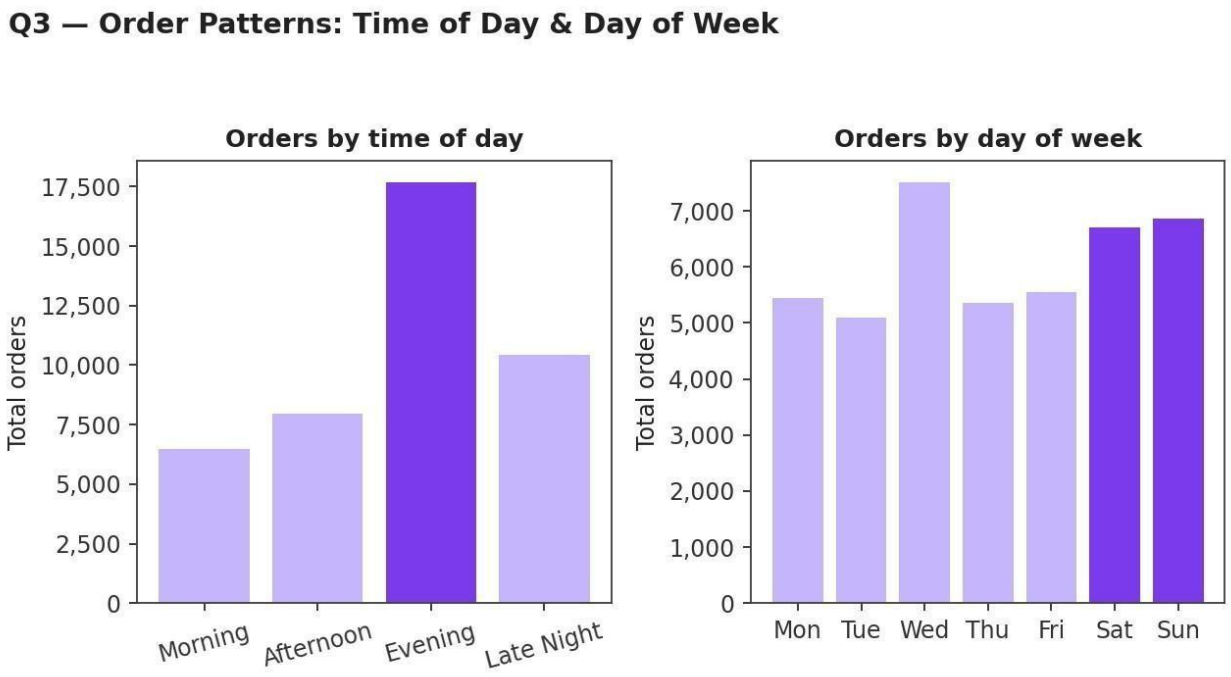
However, the average alone does not show whether delivery performance is improving or getting worse. Therefore, trend analysis is also used to observe how delivery time changes over the period.

Hauz Khas – Delhi was identified as the highest SLA-risk zone, where average delivery time increased from 11.0 minutes to 13.0 minutes during the 15-day period. This upward trend indicates a potential operational issue such as dark-store capacity limitations or rider availability problems.

Hauz Khas - Delhi's delivery-time trend is the single clearest action item in this report: a climb from 11.0 to 13.0 minutes in 15 days, if it continued at the same rate, would push the zone well past the point where customers start noticing and churning — worth investigating (rider headcount at that dark store, order packing bottlenecks, or a demand spike outmatching current staffing) before the next weekly ops review rather than after.

Q3 — Order Patterns: Time of Day & Day of Week

Left: total orders summed by time-of-day band across the window.
Right: total orders summed by day of week.



Generated with pandas (aggregation) + matplotlib (visualization) directly from zepto_order_data.csv.

- Evening (5pm-9pm) is the clear peak window, capturing 41.6% of all orders — consistent with quick-commerce apps being used heavily for dinner-time top-ups and last-minute ingredient runs.

- Weekend order volume runs 28.6% higher than the average weekday, reflecting more discretionary and social - occasion ordering (snacks, beverages) on Saturdays and Sundays compared to the routine restocking that dominates weekdays.

- Both cuts matter for very different operational decisions: time-of day drives rider shift-scheduling within a single day, while day-of week drives dark-store staffing and inventory pre-positioning decisions made a day or more in advance.

Demand seasonality refers to recurring patterns in customer demand over different periods of time. In this analysis, order patterns are studied using two time scales: time of day and day of week.

The analysis shows that evening (5 PM–9 PM) is the peak ordering period, accounting for 41.6% of all orders. This indicates that customer demand is highest during the evening, when customers

commonly place orders for dinner-time requirements, snacks, and last-minute purchases.

The analysis also shows that weekend order volume is 28.6% higher than the average weekday. This indicates increased demand on Saturdays and Sundays compared with regular weekdays.

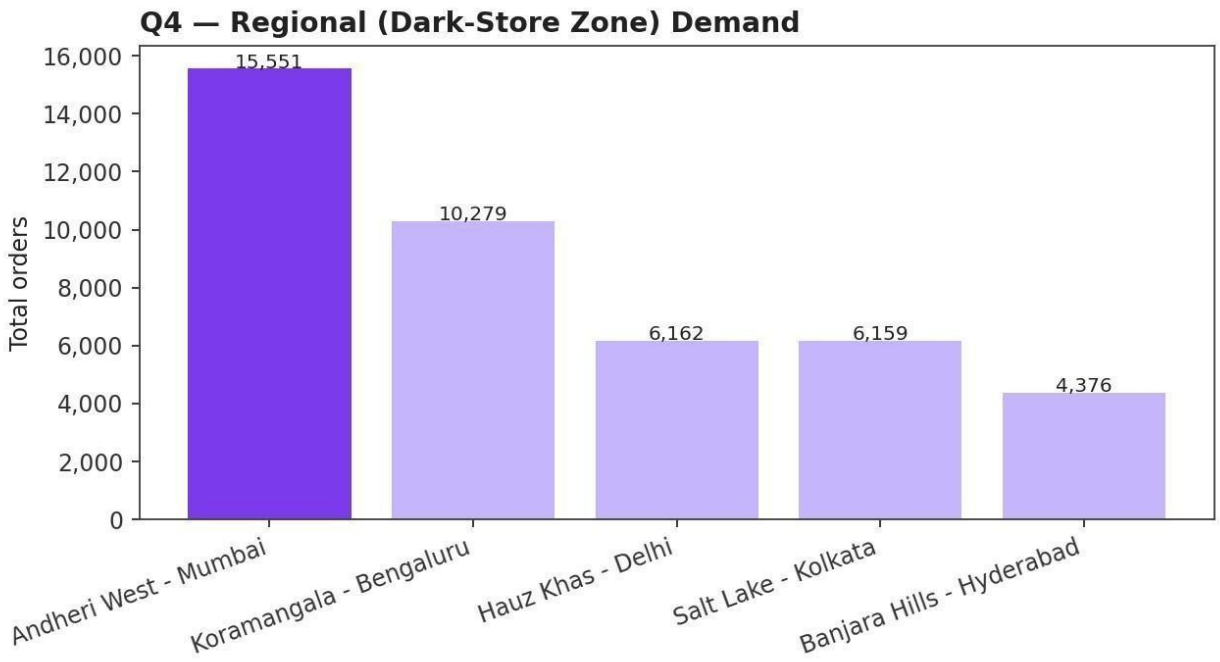
These patterns represent demand seasonality at multiple time scales. Time-of-day patterns are useful for rider scheduling and packing staff allocation, while day-of-week patterns help with inventory planning and dark-store staffing

With 41.6% of orders landing in the evening window, rider shift schedules and dark-store packing-staff rosters should be weighted toward that window rather than spread evenly across the

day — the delivery-time risk in Section 5 is also most likely to materialize during this same peak, since capacity is stretched thinnest exactly when order volume is highest.

7 Q4 — Regional (Dark-Store Zone) Demand

Total orders by dark-store zone, summed across the full 15-day window.



- Andheri West - Mumbai leads with 15,551 orders (36.6% of total demand), noticeably ahead of the other 4 zones tracked.
- Demand tapers off fairly smoothly from the top zone down — at real Zepto-scale volumes across a full city network, this concentration is typically sharper still, with a handful of dense residential/office zones accounting for the large majority of orders and newer or lower-density zones long-tailed.
- This view is what drives dark-store replenishment frequency, rider fleet allocation across zones, and new-dark-store siting decisions (a consistently high-demand zone is a candidate for a second, closer dark store to protect delivery time as order density grows).

In this analysis, total orders are compared across the 5 darkstore zones to identify areas with the highest demand.

Andheri West – Mumbai recorded the highest demand with 15,551 orders, accounting for 36.6% of total orders. It was noticeably ahead of the other zones, showing that demand is concentrated more heavily in this particular zone.

To measure this concentration formally, the analysis uses the Herfindahl-Hirschman Index (HHI). HHI is calculated by squaring the order-share percentage of each zone and adding the results. A lower HHI indicates that demand is more evenly distributed, while a higher HHI indicates that demand is concentrated in a smaller number of zones.

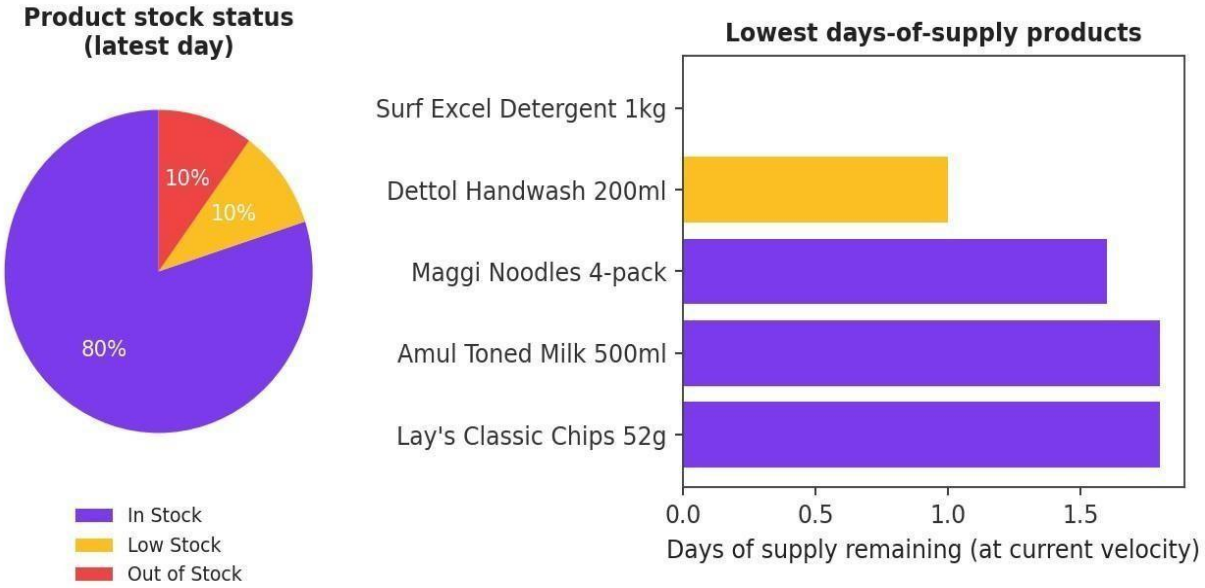
The high demand concentration in Andheri West means that any capacity problem, rider shortage, or dark-store disruption in this zone could have a significant impact on overall order fulfillment. Therefore, rider allocation, replenishment, and backup fulfillment capacity should consider the level of demand in each zone.

With Andheri West - Mumbai carrying roughly a third of demand, a capacity disruption there (a dark-store outage, a rider shortage, a local weather event) would affect total order volume far more than an equivalent disruption in the smallest zone — backup-rider pools and overflow fulfillment plans should be sized to that concentration, not spread evenly across all 5 zones.

8 Q5 — Inventory & Stockout Risk

Left: latest-day stock status across all products. Right: the five products with the fewest days of supply remaining at their current average daily order velocity.

Q5 — Inventory Risk



Generated with pandas (aggregation) + matplotlib (visualization) directly from zepto_order_data.csv.

- Of 10 products, 8 are In Stock, 1 is in the Low Stock band, and 1 is fully Out of Stock as of the latest day.

- The Out of Stock product (Surf Excel Detergent) is a slower mover, which limits the order-volume impact — the same event on the #1 best-seller from Section 4 would be a materially bigger problem for both revenue and the habitual-restocking use case that keeps customers on the app.

- "Days of supply" (current stock ÷ average daily orders) is a more actionable cut than a static stock count: it converts inventory into a time-to-stockout estimate per product, which is what actually triggers a dark-store replenishment run.

The analysis found that out of 10 products, 8 were In Stock, 1 was in Low Stock, and 1 was Out of Stock on the latest day. The Out of Stock product was Surf Excel Detergent.

Days of Supply is more useful than looking only at the current stock count because it converts inventory into an estimated time until stockout. This helps identify which products need replenishment sooner.

The analysis also uses the concept of a Reorder Point, which is the inventory level at which a new replenishment should be triggered. It is calculated using average daily demand, replenishment lead time, and safety stock.

Overall, 2 out of 10 products were at Low Stock or below, showing that inventory monitoring and timely replenishment are important for preventing stockouts.

2 of 10 products are already at Low Stock or below in this short window — at real order volume, the same days-of-supply calculation run continuously per product per dark store is what would feed an automated replenishment queue, rather than a store associate noticing a stockout after it has already cost sales and disappointed a customer expecting 10-minute delivery.

9 Summary of Key Statistics

Metric	Value (this window)
Average delivery time	10.7 min (SLA target: 10 min)
Product-days exceeding SLA	40.0%
Top single-window seller	Amul Toned Milk 500ml (9,004 orders)
Top seller's share of total orders	21.2%
Peak ordering window	Evening (5pm-9pm) (41.6% of orders)
Weekend vs. weekday order lift	+28.6%
Leading dark-store zone	Andheri West - Mumbai (36.6% of demand)
Products at Low Stock or below	2 of 10
Highest SLA-risk zone	Hauz Khas - Delhi (11.0 → 13.0 min)

10 Next Steps: Moving From Descriptive → Predictive

Delivery-time forecasting	**Predicts per-zone delivery time ahead of a shift, so rider allocation can be adjusted before an SLA breach happens, not after**
Demand forecasting	**Predicts per-product/per-zone order volume to set dark-store reorder points automatically instead of reacting to a Low Stock alert**
Anomaly detection	**Flags a sudden delivery-time spike or unexpected demand surge at a specific dark store automatically, before a human notices the trend**

11. Reference page

1. The zepto dataset -

• Synthetic dataset from Claude AI.

2. Pandas documentation –

- **Data cleaning**
- **Data manipulation and analysis.**

3. Matplotlib documentation-

- **Creating graph**
- **visualization.**